

Iron Sentinel: Internal Network Vulnerability Scan

Vulnerability Assessment Report

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1. Summary

An internal network vulnerability scan was performed against the Iron Sentinel Lab's Windows Server 2022 host using Nessus Essentials. The scan was run without authentication credentials, which limits visibility into installed software versions and missing patches but still identifies network facing configuration issues.

The scan returned 26 total findings. One medium severity configuration issue and 25 informational items describing host and service characteristics. No Critical or High severity vulnerabilities were identified.

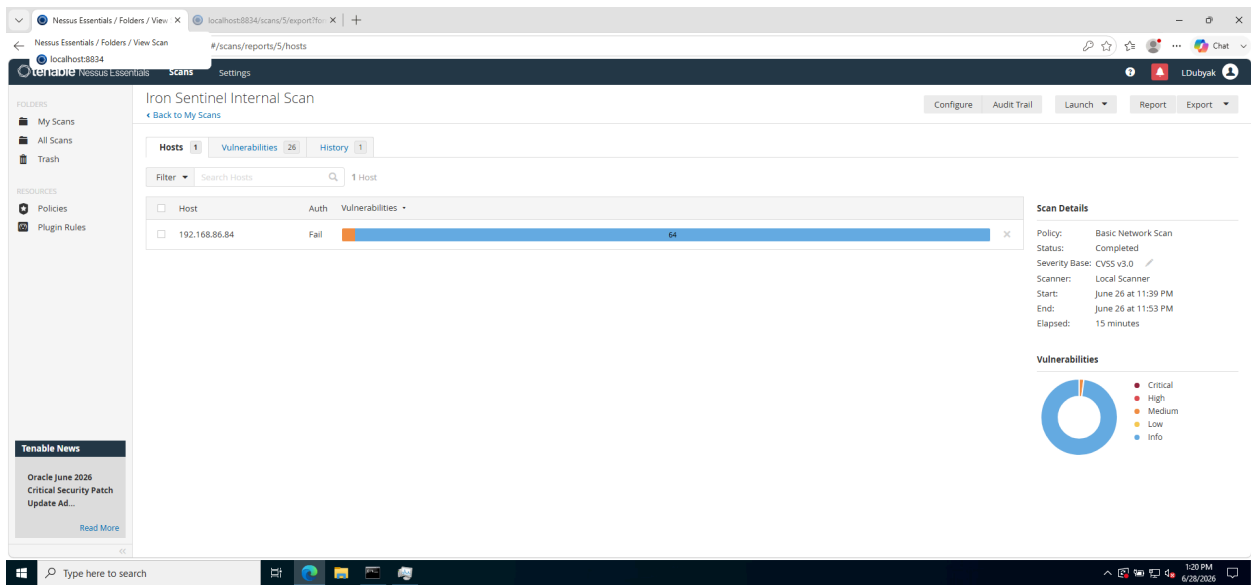
Target Host	192.168.86.84 (Windows Server 2022)
Scan Type	Unauthenticated Basic Network Scan
Scanner	Nessus Essentials, local scanner
Duration	15 minutes
Total Findings	26 (1 Medium, 25 Informational)
Severity Base	CVSS v3.0

2. Findings

Severity	CVSS	Finding	Port	Service
Medium	5.3	SMB Signing not required	445	microsoft-ds

SMB signing is not enforced on the host. Without signing, an attacker positioned on the network could intercept and tamper with SMB traffic,

creating a path for man-in-the-middle attacks against file sharing and authentication exchanges.



Recommendation

Enforce SMB signing via Group Policy: Computer Configuration > Windows Settings > Security Settings > Local Policies > Security Policies > “Microsoft network server: Digitally sign communications (always).” This setting should be validated in a test environment before production rollout because it can affect compatibility with legacy SMB clients.

Informational Findings

The remaining 25 findings are informational and describe normal host and service characteristics rather than security weaknesses. They are grouped below by category for reference.

Category	Count	Examples
Service Detection	8	DCE Services Enumeration, Service

		Detection, WS-Management, Server Detection
Protocol Notices	10	SSL/TLS (multiple issues), TLS Version 1.2 Detection, Post-Quantum Cipher Notice
Host identification	5	OS identification, CPE, Device Type, FQDN Resolution
Scan metadata	2	Nessus Scan Information, Credential Status (no credentials provided)

3. Scan Limitations

- This was an unauthenticated network scan. No Windows credentials were provided, so Nessus could not perform a full OS patch-level assessment or enumerate installed software versions.
- Only one host (192.168.86.84) was in scope, consistent with the single-host Iron Sentinel lab environment.
- Results reflect the state of the host at scan time; no historical data is available since this was a single scan.

4. Recommended Next Steps

- Apply the SMB signing remediation above and re-scan to confirm the finding clears.
- Run a follow-up authenticated scan using local administrator credentials to assess patch levels and installed software. This is the standard next step in a production vulnerability management program.

- Establish a recurring scan cadence such as monthly to track vulnerability trends over time rather than relying on single point-in-time snapshots.